ABC SENIOR SECONDARY SCHOOL

CBSE CLASS XII — BOARD EXAMINATION (THEORY)

Subject: Science

Time Allowed: 3 Hours | Maximum Marks: 80 | 29 May 2026

Name: __________________________ Roll No.: __________________________
Class/Section: _____________

GENERAL INSTRUCTIONS
1. All questions are compulsory.
2. Read each question carefully before answering.
3. Figures to the right indicate full marks.
4. Use of calculator is not permitted unless stated.

SECTION A — MULTIPLE CHOICE QUESTIONS

Each question carries 1 mark. Select the correct option.

Q1. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q2. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q3. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q4. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q5. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q6. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q7. (1 marks)

A convex mirror always forms an erect, virtual image regardless of the object'th position relative to it. Given this information and considering a concave lens with focal length 'f', which option correctly describes what happens when parallel rays enter such a lens?

(a) {'text': 'The image formed is real, inverted and diminished.'}

(b) {'text': "The image remains virtual but becomes erect and magnified as the object moves closer to the lens than its focal length 'f'."}

Q8. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q9. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q10. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q11. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q12. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q13. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q14. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q15. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q16. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q17. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q18. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q19. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

Q20. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

SECTION B — VERY SHORT ANSWER

Answer in 40-60 words. Each question carries 2 marks.

Q21. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

Q22. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

Q23. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

Q24. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

Q25. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

Q26. (2 marks)

Explain an important concept from the syllabus related to B in the context of the uploaded material. (2 marks)

SECTION C — SHORT ANSWER

Answer in 60-80 words. Each question carries 3 marks.

Q27. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

Q28. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

Q29. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

Q30. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

Q31. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

Q32. (3 marks)

Explain an important concept from the syllabus related to C in the context of the uploaded material. (3 marks)

SECTION D — CASE BASED / SOURCE BASED

Read the passage and answer the sub-questions. Each question carries 4 marks.

Q33. (4 marks)

Explain an important concept from the syllabus related to D in the context of the uploaded material. (4 marks)

Q34. (4 marks)

Explain an important concept from the syllabus related to D in the context of the uploaded material. (4 marks)

Q35. (4 marks)

A circuit consists of various components connected by wires, including resistors and capacitors as shown in Figure [fig_0007]. The battery provides a potential difference between its terminals. When the switch is 'ON', current flows through this closed loop.

Figure

(a) Explain how Kirchhoff'sintegration rule can be applied to calculate the currents flowing through each resistor in Figure [fig_0007] when the switch is 'ON'. Assume all resistors have different resistance values and that there are no other power sources or complex components within this circuit?

(b) Given a potential difference of V volts across the battery terminals in Figure [fig_0007], calculate the total current flowing through the entire closed loop when all switches are 'ON'. Assume ideal wires with no resistance and that each resistor has its own labeled value?

(2) If one of the resistors in Figure [fig_0007] is removed while keeping all other components intact (and switches 'ON'), describe qualitatively how this would affect the current distribution and voltage across each remaining resistor. Assume ideal conditions?

(3) Determine whether it's possible to have a situation in Figure [fig_0007] where no current flows through one of the resistors while all switches are 'ON'. Explain your reasoning based on Kirchhoff'sintegration rule?

SECTION E — LONG ANSWER

Answer in detail with diagrams where necessary.

Q36. (5 marks)

Explain an important concept from the syllabus related to E in the context of the uploaded material. (5 marks)

Q37. (5 marks)

A caterpillar's metamorphosis is a fascinating process that involves several stages. Explain how the physical changes in an insect during its life cycle contribute to this transformation, and discuss why these processes are vital for survival from both ecological (ecosystem) and evolutionary perspectives.

(a) What is the function of a leaf in an insect's life cycle?
(b) How does molting contribute to growth, particularly during metamorphosis?

Based on Figure 0012 (not provided), identify and explain the role of each component in an insect's life cycle. Discuss how these stages are interconnected, considering both ecological balance and evolutionary advantages.

Considering Figure 0014 (not provided), analyze the energy dynamics during metamorphosis in insects. How does this process impact their survival, reproduction rates, and population control within an ecosystem?

Using Figure 0015 (not provided), explain the role of environmental factors such as temperature and humidity in triggering metamorphosis. How do these conditions affect survival rates during different life cycle stages?

Study the figure below.

SECTION F — EXTENDED RESPONSE

Answer with proper derivation / explanation. (8 marks)

Q38. (8 marks)

Explain an important concept from the syllabus related to F in the context of the uploaded material. (8 marks)